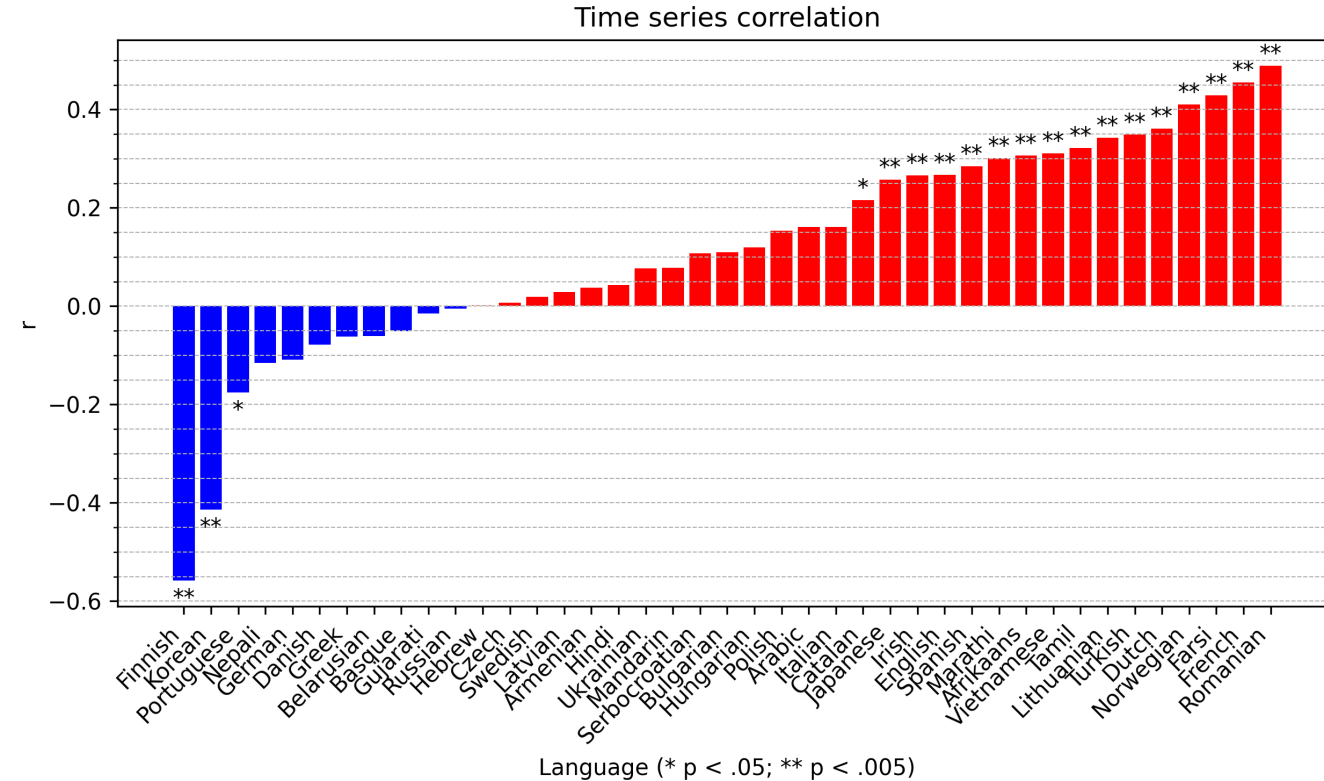


MLM – Alice project

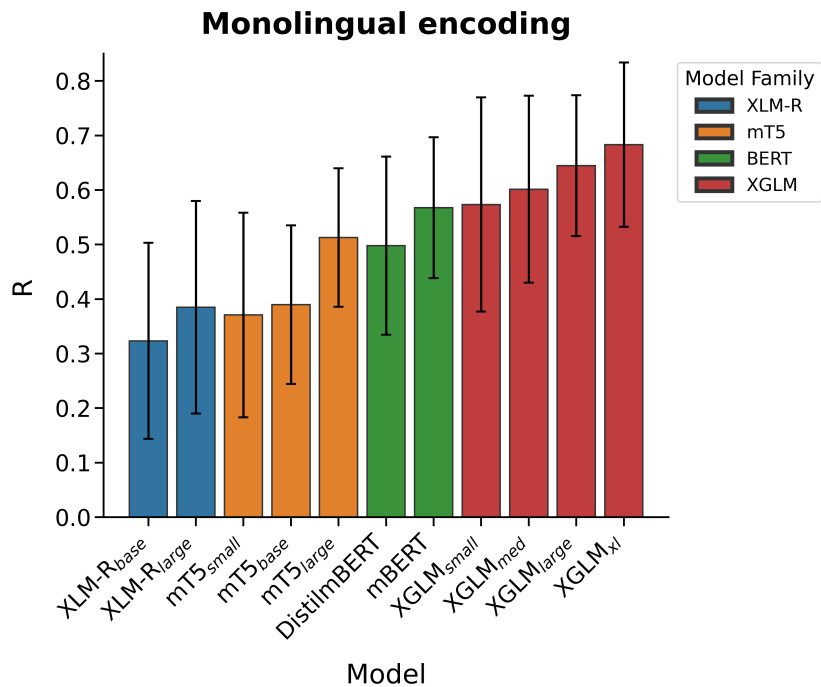
November 17 meeting

Time series correlation



- Correlations are significant in 16 languages
- Irish is not supported by whisper-timestamped
→ 15 languages

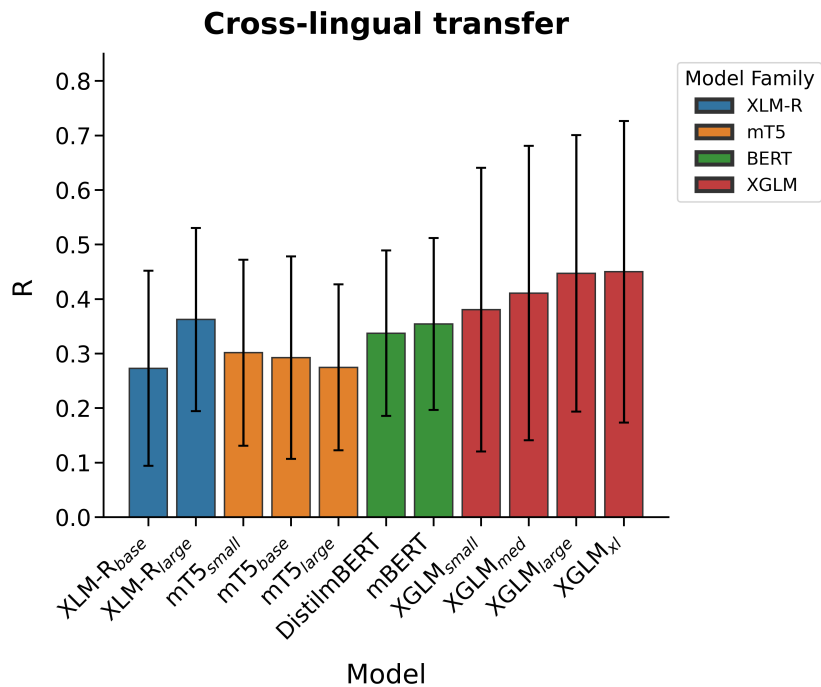
Encoding performance by model



Monolingual encoding

- Best layer for each model
- 10-fold cross-validation (random split)
- No “embedding lag”
- R values averaged across folds and languages
- Error bars: SD across languages

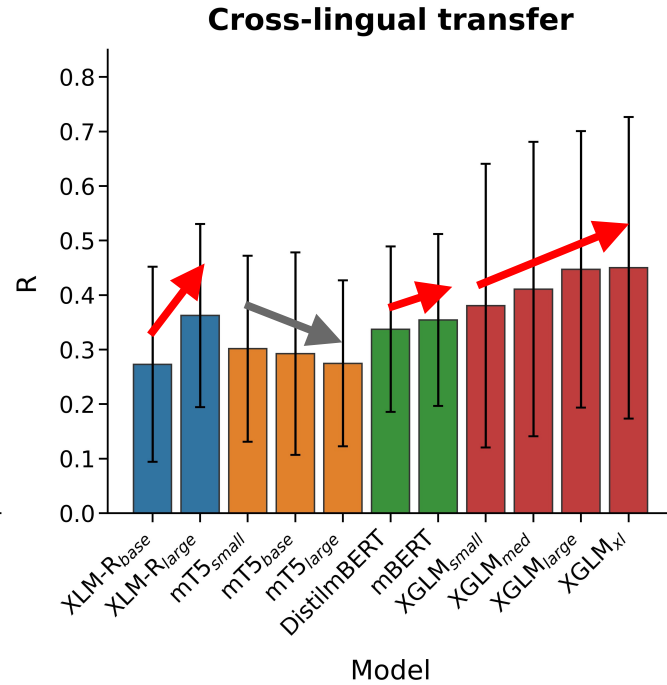
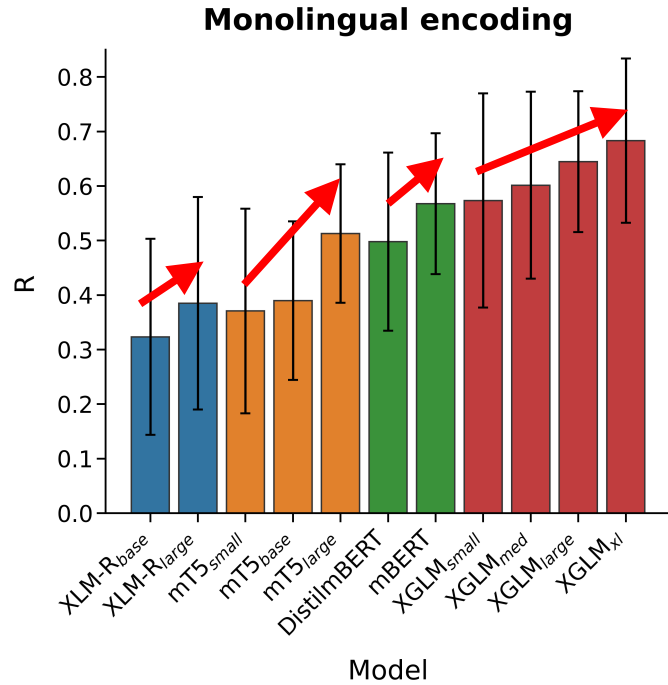
Encoding performance by model



Cross-lingual transfer

- Best layer for each model
- 10-fold cross-validation (random split)
- No “embedding lag”
- R values averaged across languages (there are no folds)
- Error bars: SD across languages

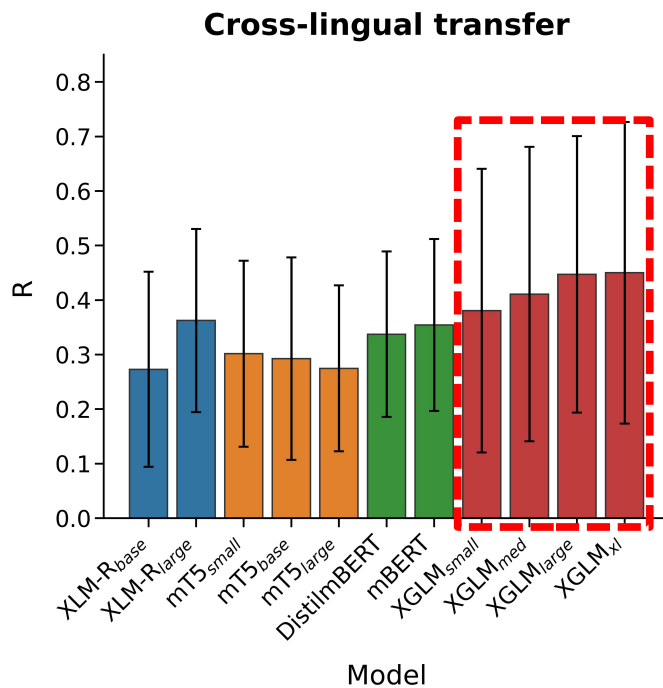
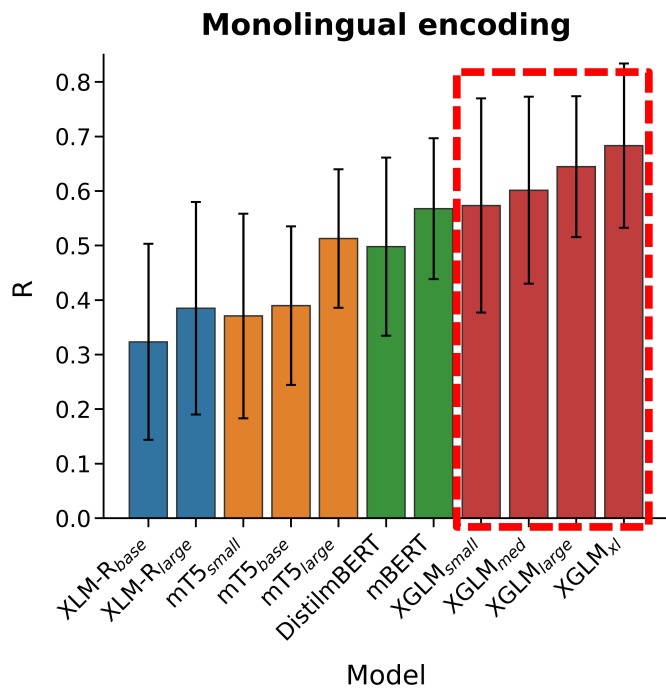
Model capacity



Larger-capacity models outperform smaller models (except for mT5 in cross-lingual transfer)

Cfr. Schrimpf et al. 2021

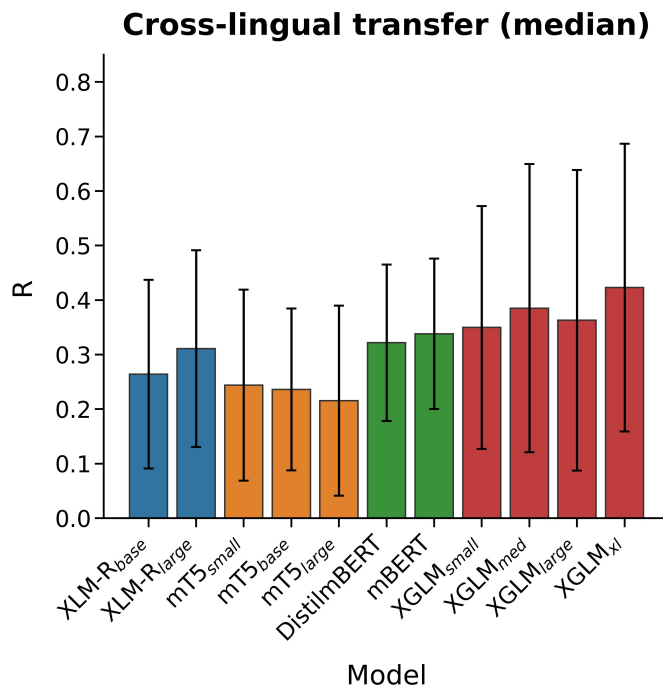
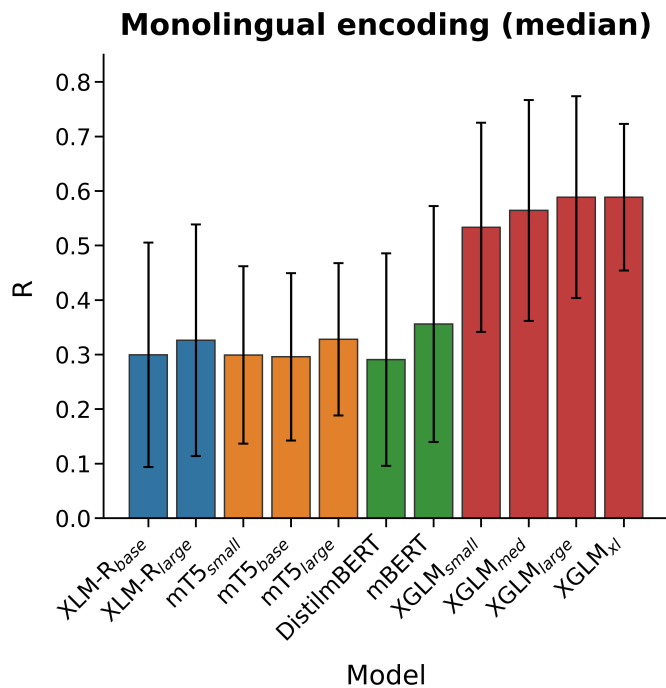
Auto-regressive vs. bidirectional



Auto-regressive
language models
outperform the
bidirectional ones

Cfr. Schrimpf et al. 2021

Median layer

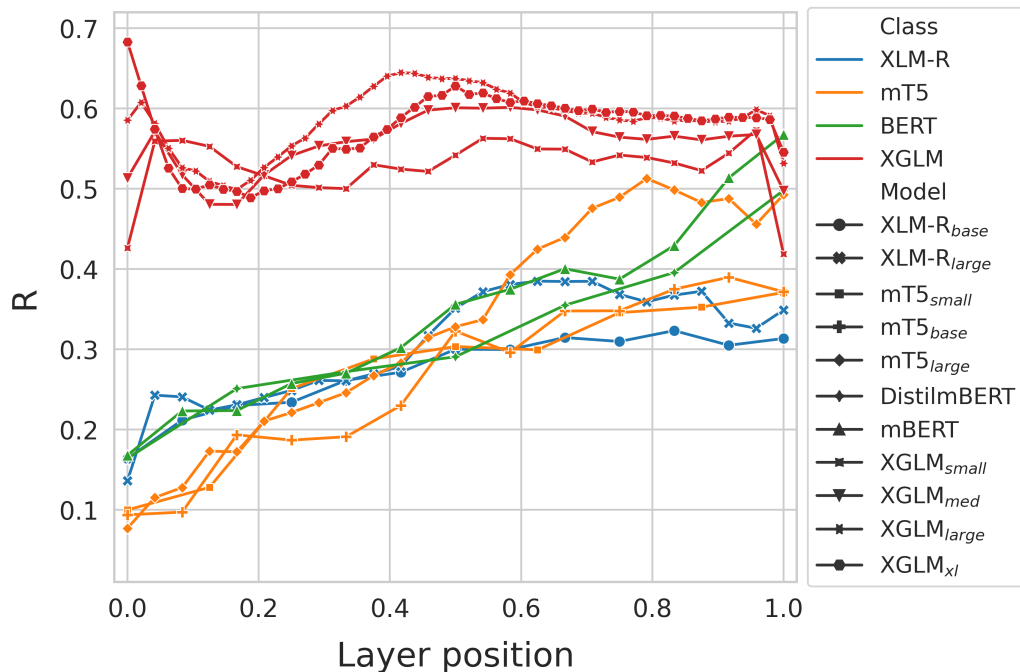


The results are similar when considering the median (as opposed to best) layer

- Bigger > smaller
- Causal > bidirectional

Encoding performance by layer

Monolingual encoding by layer

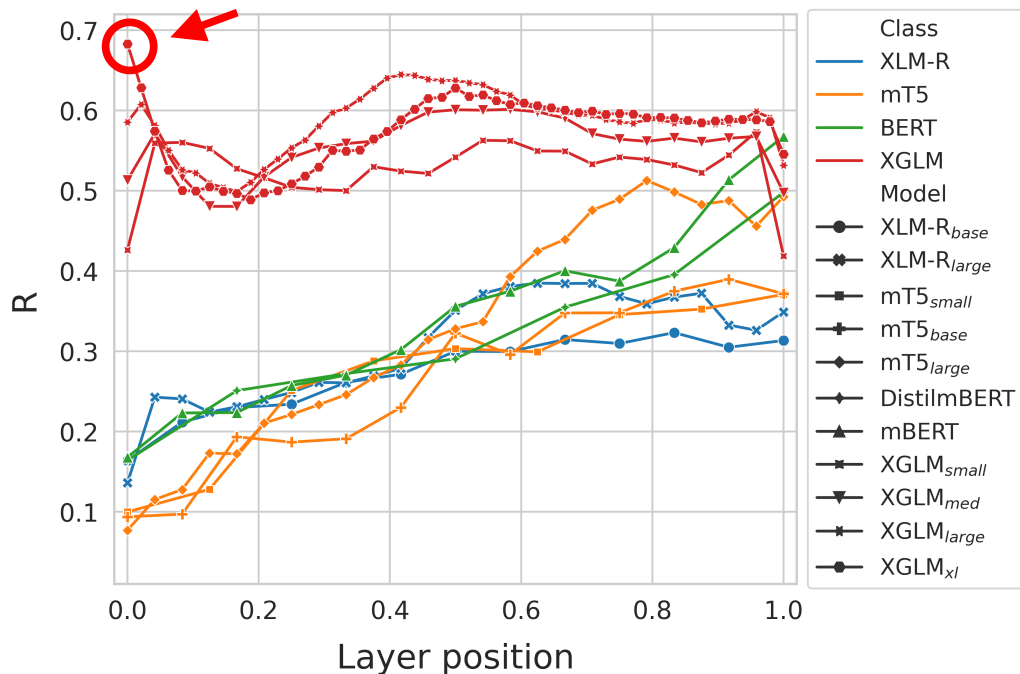


Monolingual encoding

- Middle-to-deep layers achieve the strongest predictivity
- Intermediate layers work better for the auto-regressive models
- Deep layers work better for the bidirectional models

Encoding performance by layer

Monolingual encoding by layer

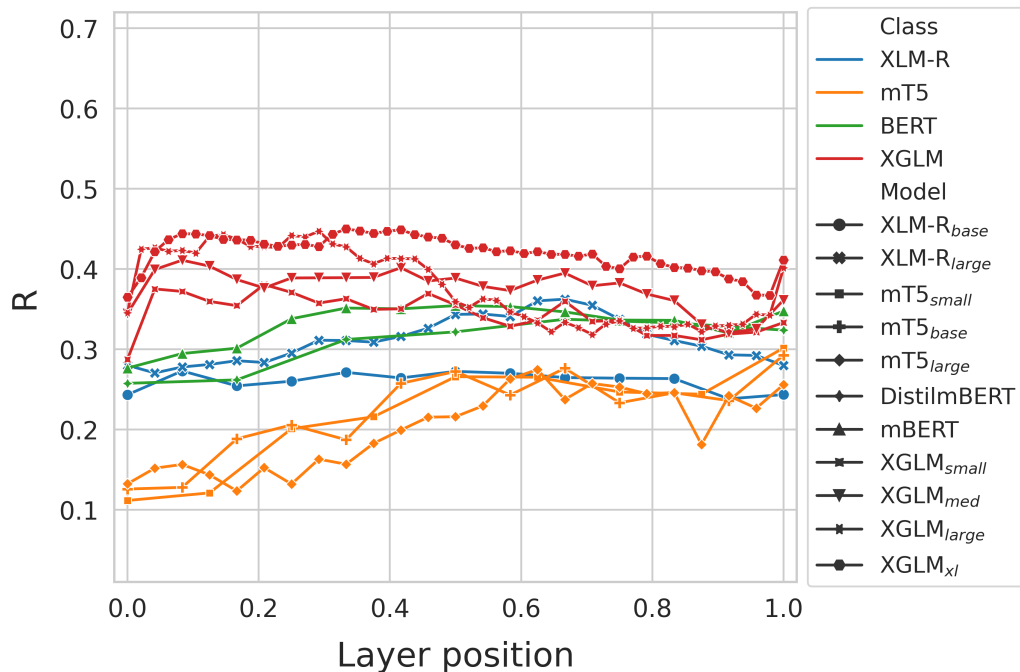


Monolingual encoding

- Middle-to-deep layers achieve the strongest predictivity
- XGLM_{xl} achieves the strongest predictivity in the embedding layer
- Deep layers work better for the bidirectional models

Encoding performance by layer

Cross-lingual transfer by layer



Multilingual encoding

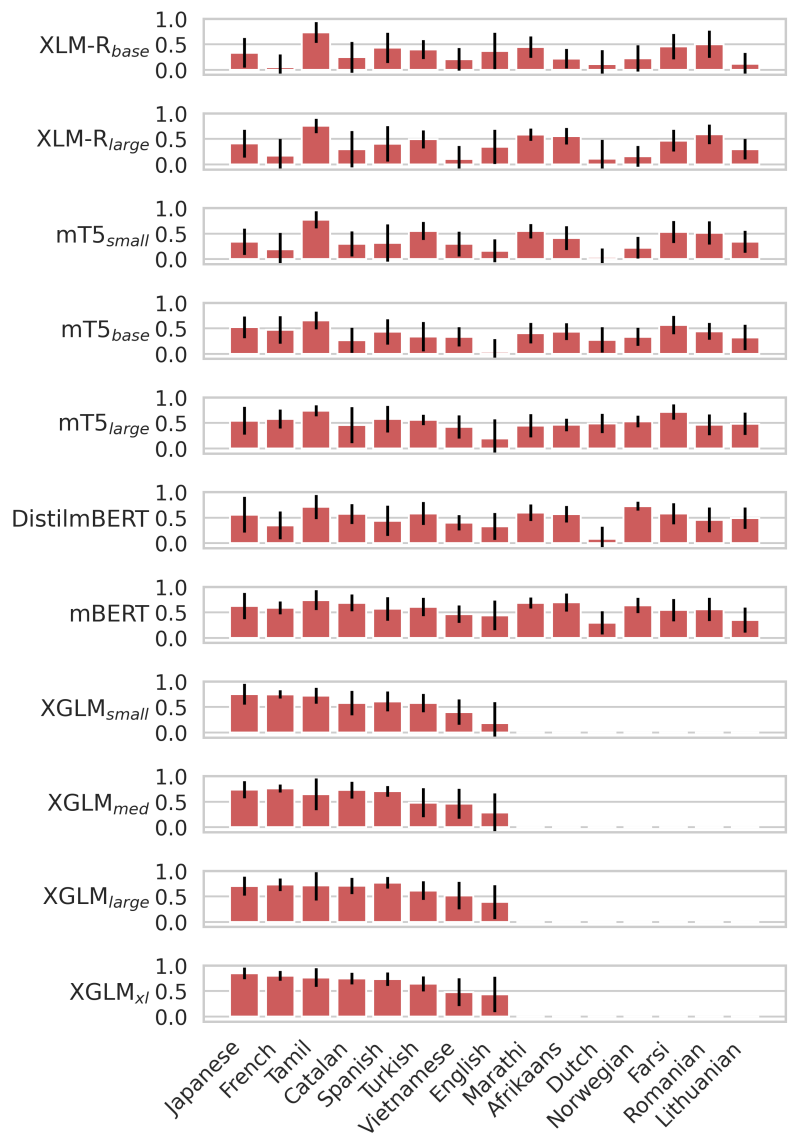
- Intermediate layers achieve the strongest predictivity
 - Early-intermediate layers work better for the auto-regressive models
 - Intermediate-deep layers work better for the bidirectional models

Monolingual encoding

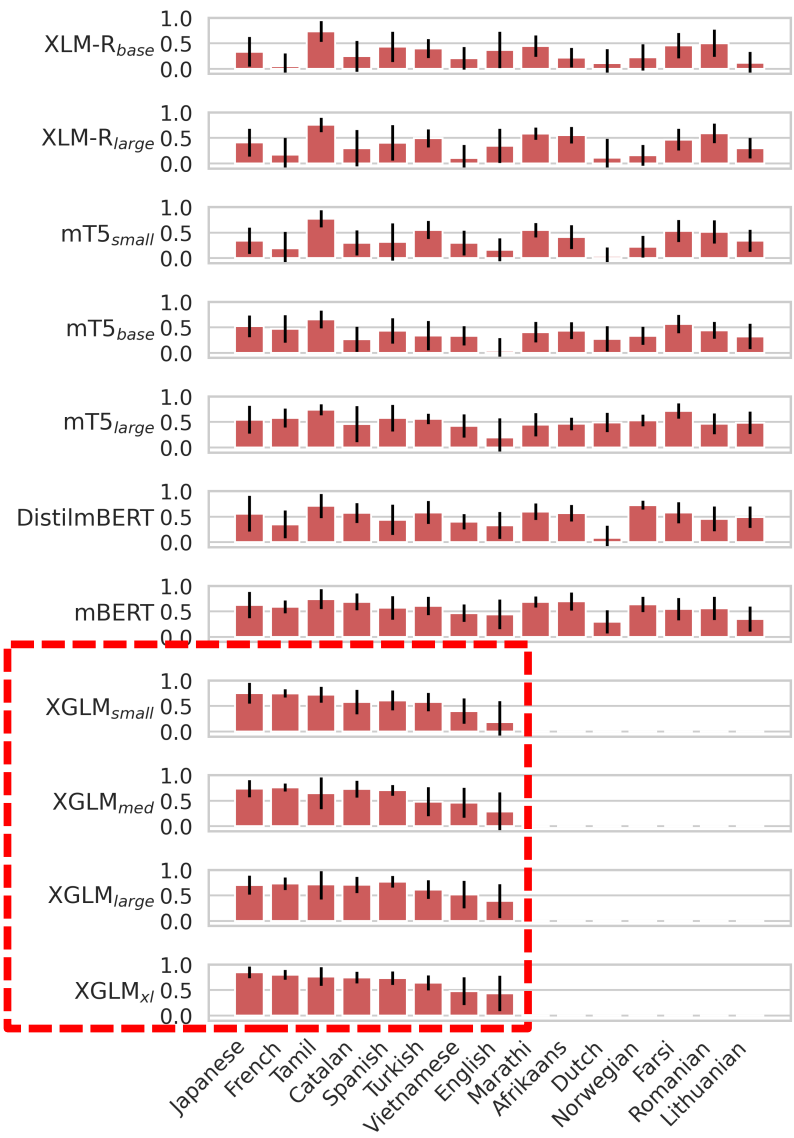
Performance by language

Monolingual encoding

- Performance obtained with the best layer (overall)
- Error bars indicate the SD across folds



Monolingual encoding

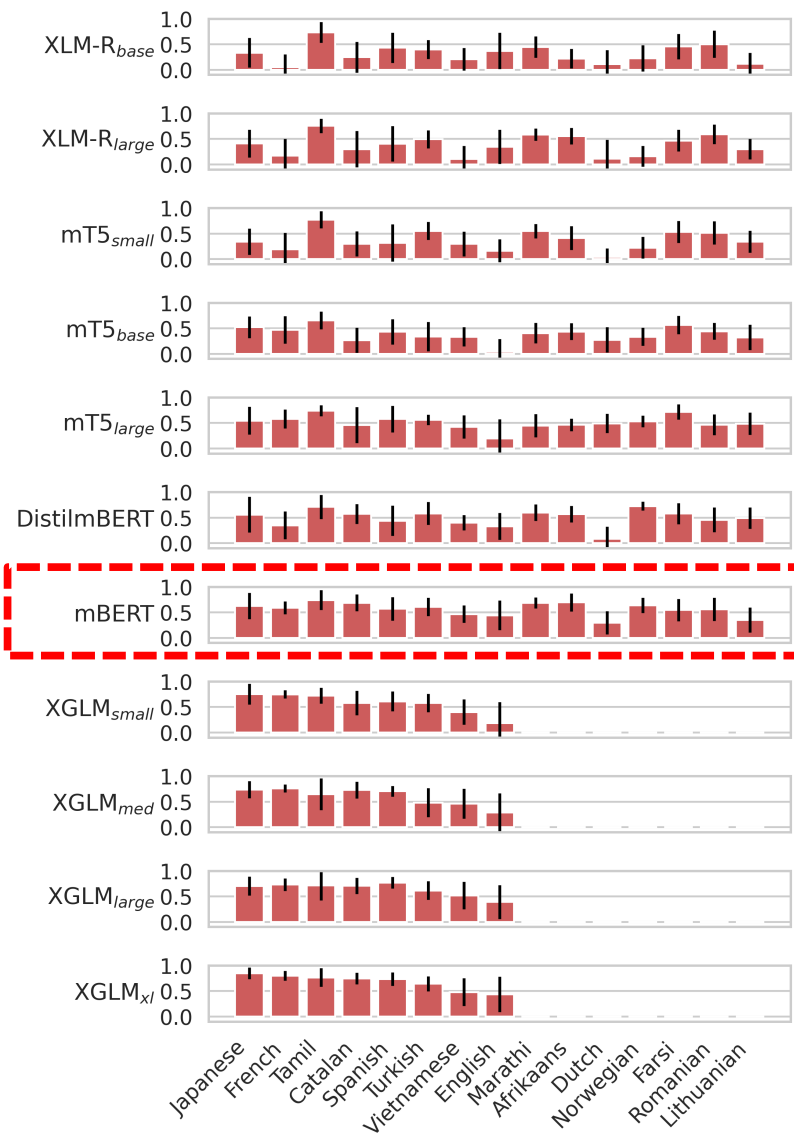


Performance by language

Monolingual encoding

- Performance obtained with the best layer (overall)
- Error bars indicate the SD across folds
- Auto-regressive models obtain a relatively strong performance across all 8 languages

Monolingual encoding

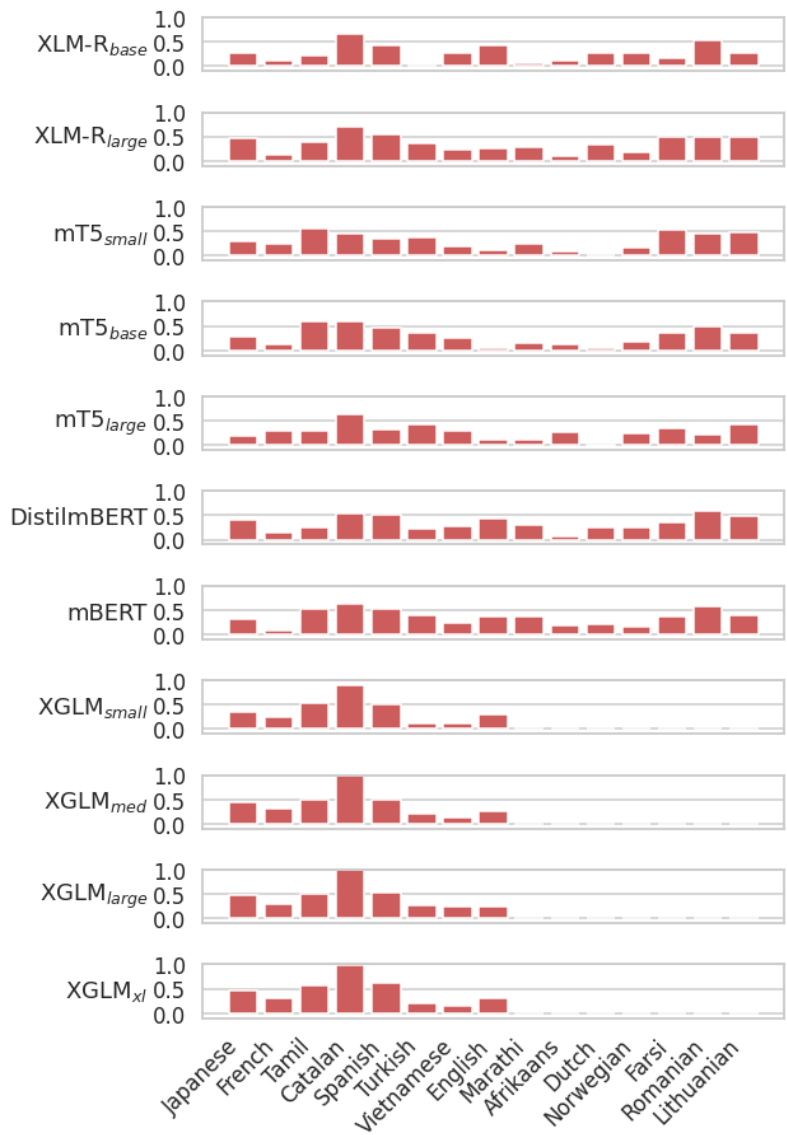


Performance by language

Monolingual encoding

- Performance obtained with the best layer (overall)
- Error bars indicate the SD across folds
- Auto-regressive models obtain a relatively strong performance across all 8 languages
- mBERT achieves relatively strong performance scores across all 15 languages

Cross-lingual transfer

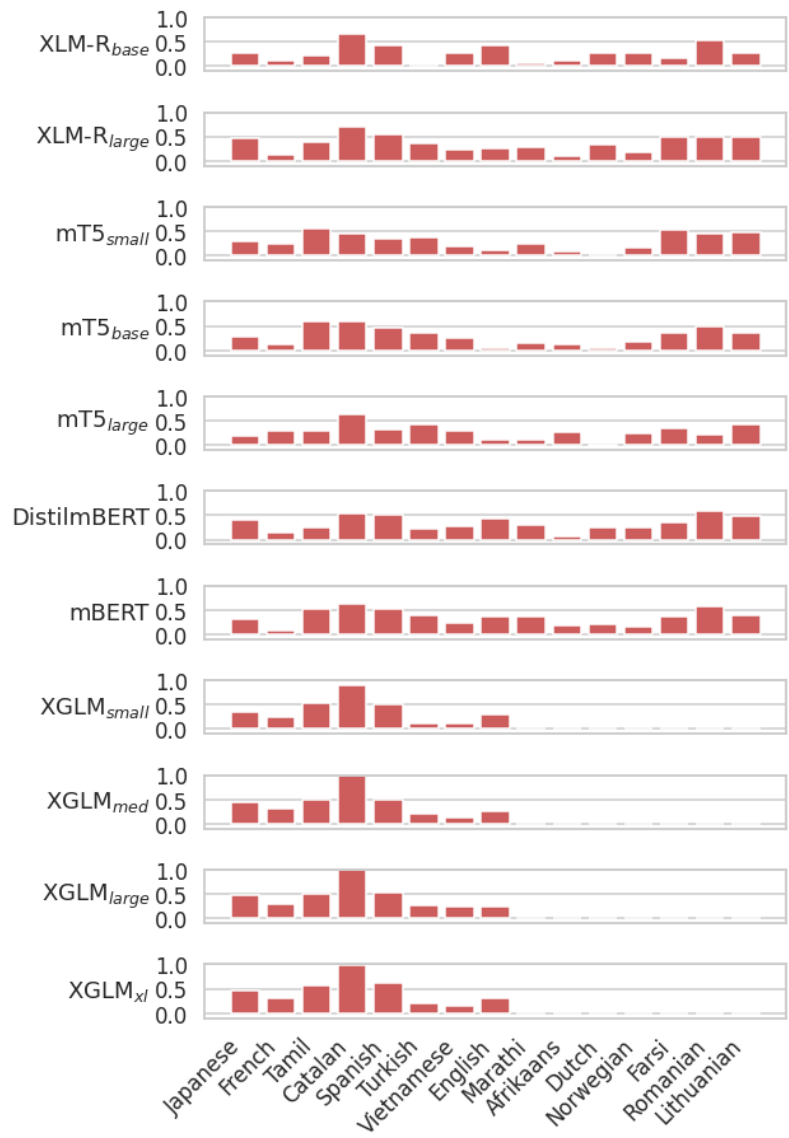


Performance by language

Multilingual encoding

For each language $l \in L$, an encoding model is fitted in all languages m in L excluding l ($m \in L \setminus \{l\}$), and tested on l .

Cross-lingual transfer



Performance by language

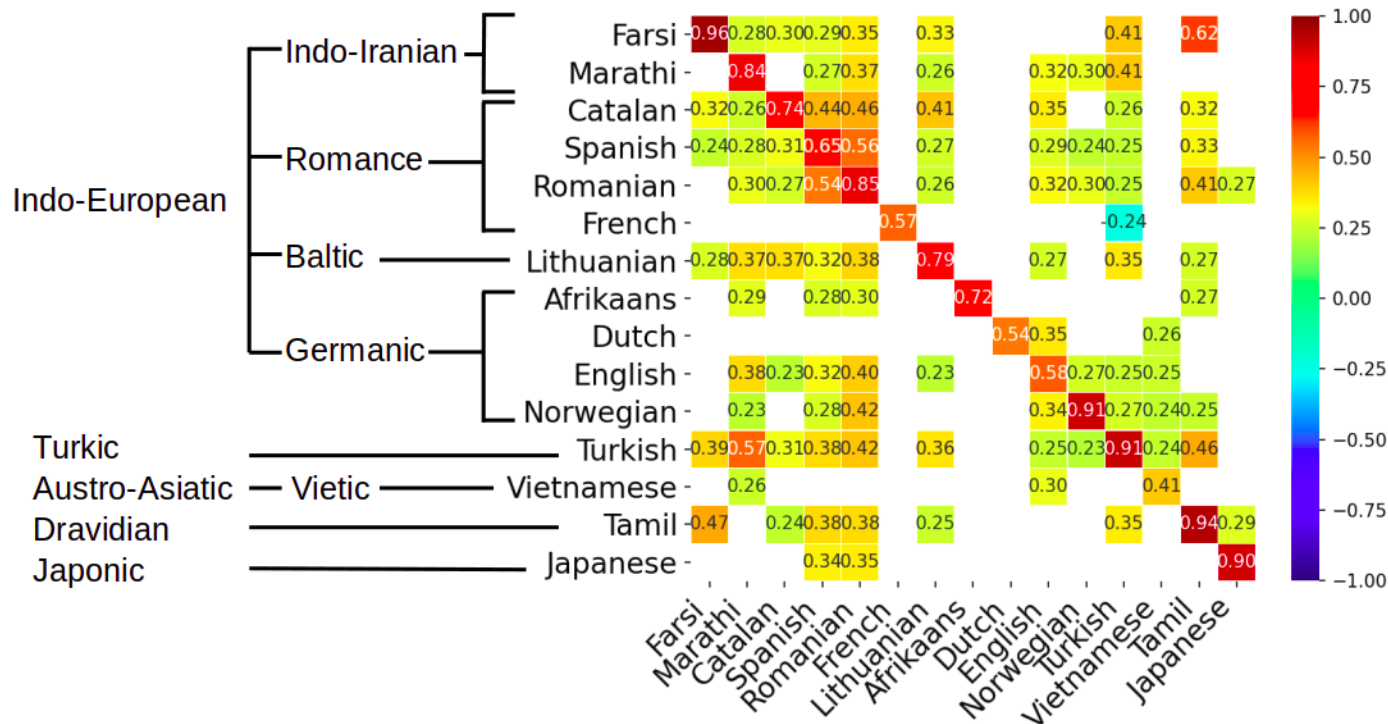
Multilingual encoding

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Alternative:

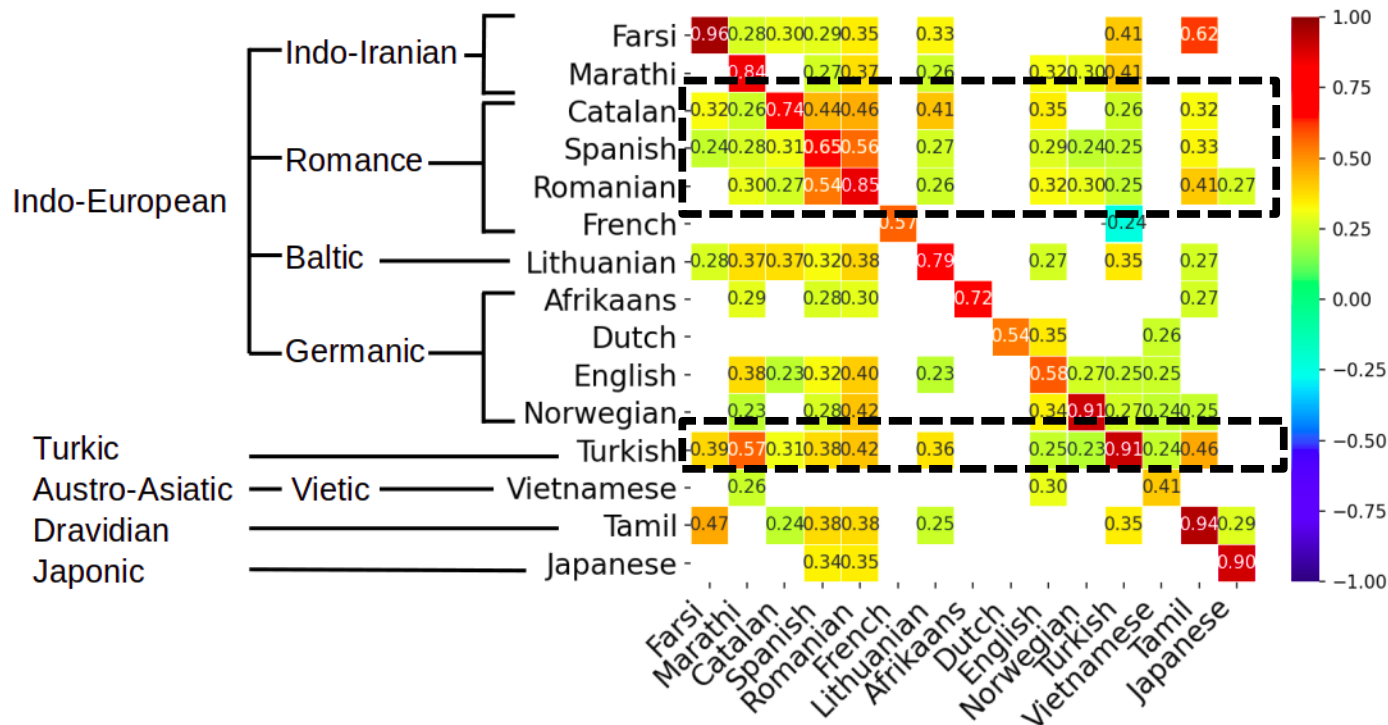
- Fit an encoding model in each language
- Transfer it zero-shot to all the other languages

Single training language



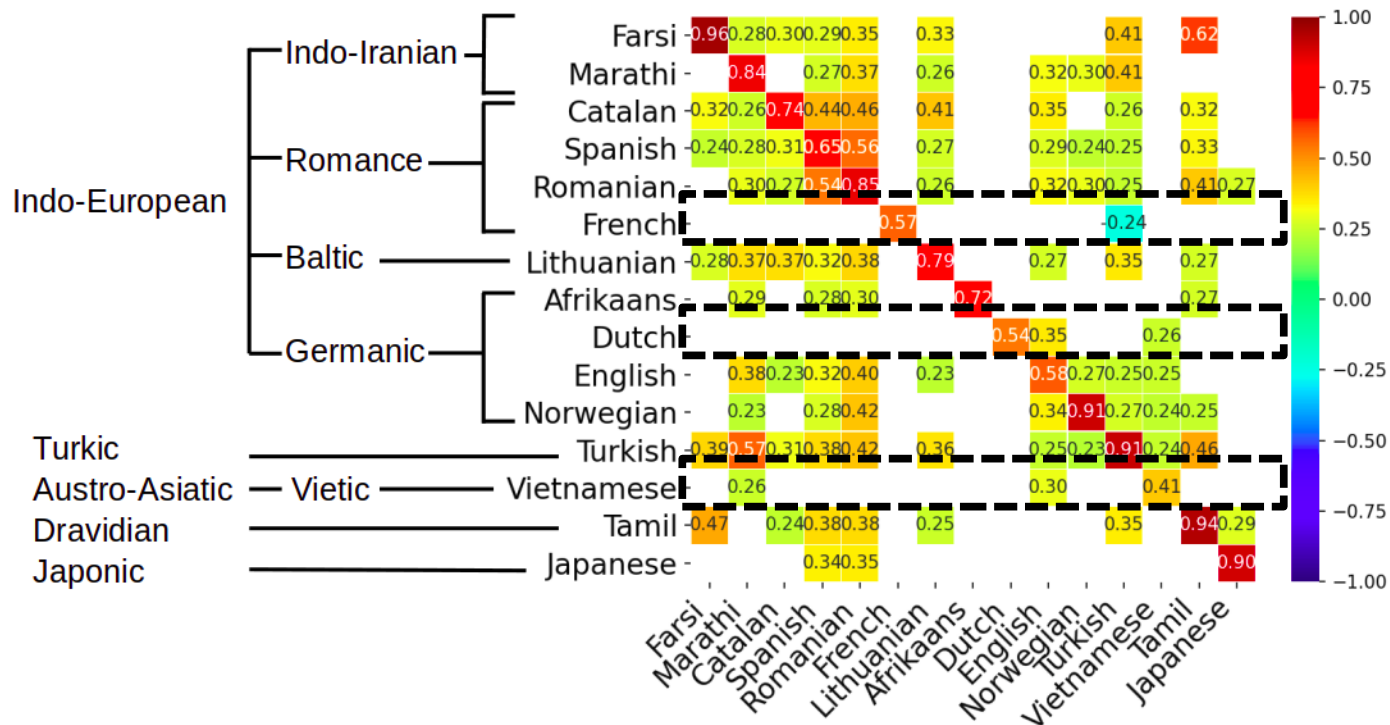
- Rows: how well a model trained in a language transfers to the other
- No clear typological pattern

Single training language



- Rows: how well a model trained in a language transfers to the other
- No clear typological pattern
- Some languages generally support well the transfer
 - Turkish, Catalan, Spanish, Romanian

Single training language

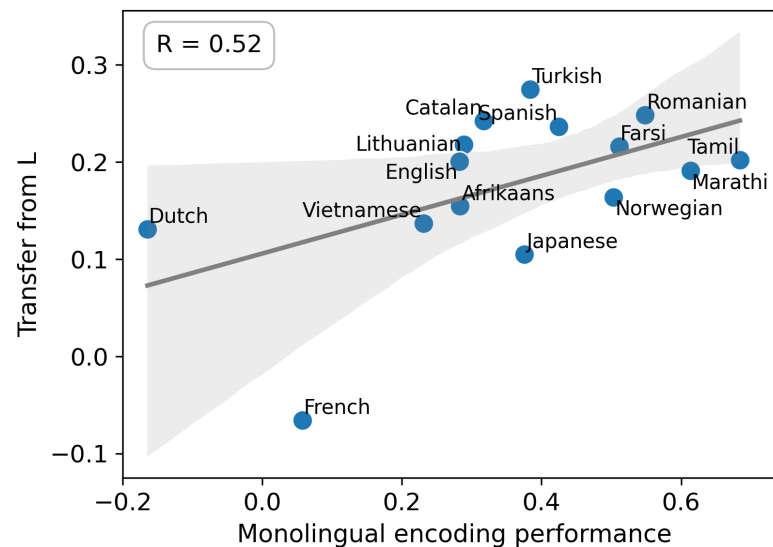


- Rows: how well a model trained in a language transfers to the other
- No clear typological pattern
- Some languages generally support well the transfer
 - Turkish, Catalan, Spanish, Romanian
- Other languages don't
 - French, Dutch, Vietnamese

Encoding and transfer performance

Transfer is generally better for some languages than for others.

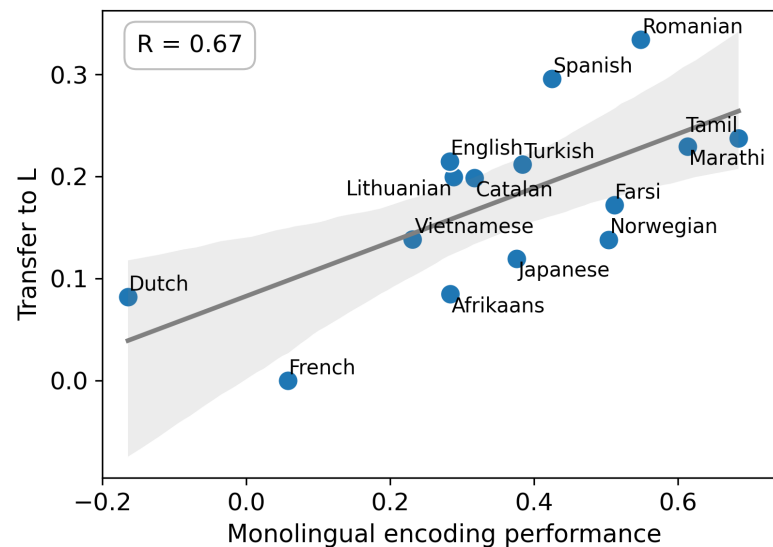
- Relationship between monolingual encoding score in a given language l and:
 - **How well the models trained on l transfers to other languages**
 - How well the models trained on other languages transfer to l



Encoding and transfer performance

Transfer is generally better for some languages than for others.

- Relationship between monolingual encoding score in a given language I and:
 - How well the models trained on I transfers to other languages
 - **How well the models trained on other languages transfer to I**



Encoding and transfer performance

Transfer is generally better for some languages than for others

```
graph TD; A[Transfer is generally better for some languages than for others] --> B[Quality of the embeddings]; A --> C[Quality of the fMRI signal];
```

The diagram consists of a central blue rectangular box with a dark blue border containing the text 'Transfer is generally better for some languages than for others'. Two dark blue arrows originate from the bottom corners of this box. The arrow on the left points downwards and slightly to the left, ending above the text 'Quality of the embeddings'. The arrow on the right points downwards and slightly to the right, ending above the text 'Quality of the fMRI signal'.

Quality of the embeddings

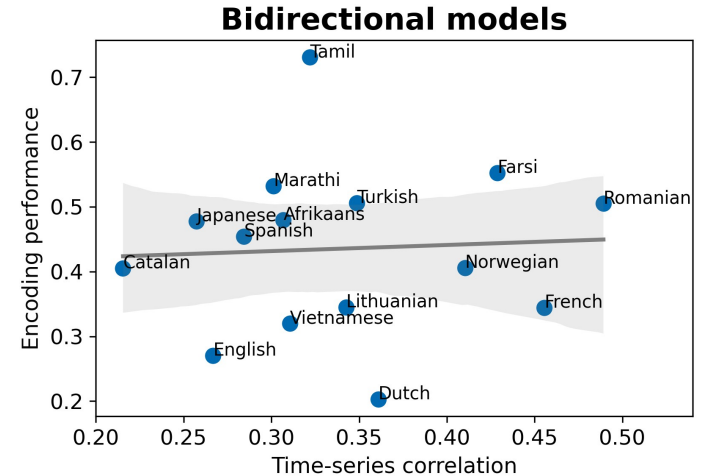
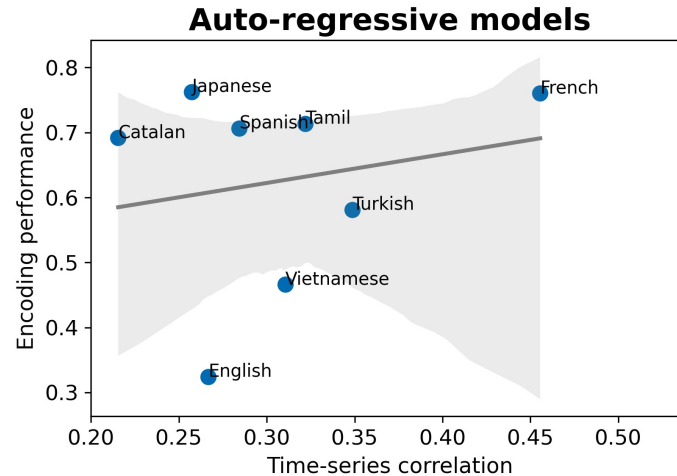
Quality of the fMRI signal

Encoding and transfer performance

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Quality of the embeddings

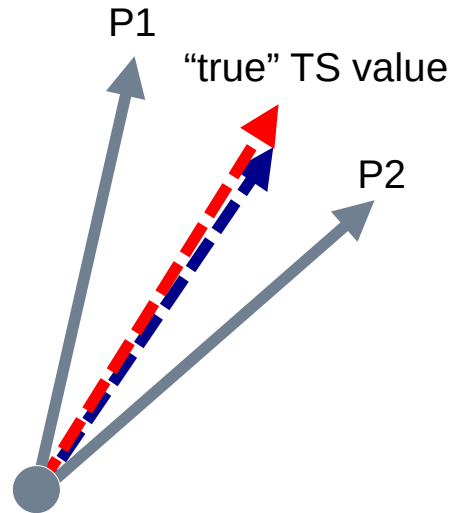
Quality of the fMRI signal



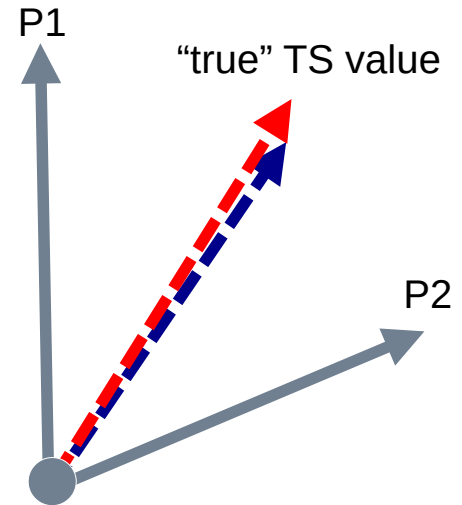
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Quality of the embeddings



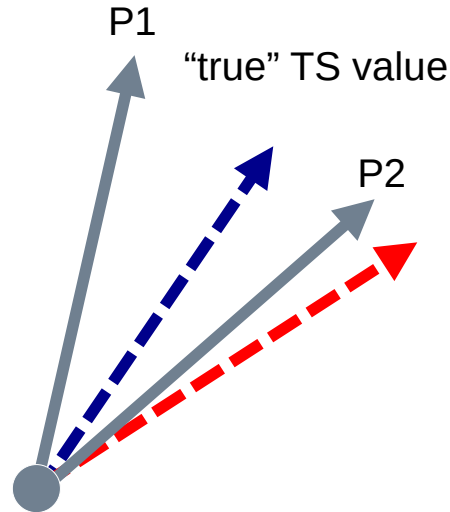
Quality of the fMRI signal



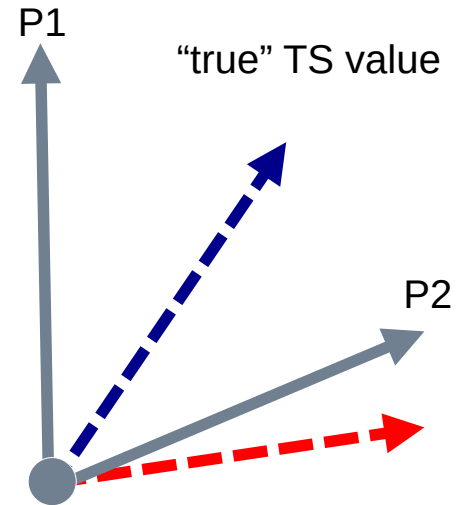
Encoding and transfer performance

Transfer is generally better for some languages than for others

Quality of the embeddings



Quality of the fMRI signal



Open questions

How to evaluate statistical significance

- Paired samples permutation test?
 - Baseline where the dependent variable is randomly shuffled
 - Compare the encoding performance between target and randomized model by doing fold-wise permutation

	Model	Shuffled
Fold 1	0.46	0.10
Fold 2	0.31	0.02
Fold 3	0.28	0.04
Fold 4	0.40	0.01
Fold 5	0.63	0.15
Fold 6	0.11	0.12
Fold 7	0.55	0.03
Fold 8	0.42	0.19
Fold 9	0.64	0.00
Fold 10	0.08	0.07

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$$p = | \Delta > \Delta_0 | / N_{\text{perm}}$$

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Open questions

Extending the drive/suppress framework

- Do we still want to do it?
 - The models are already transferred to new languages, stimuli, and participants
- If we want to do it, what do we want to test?
 - Surprisal, grammaticality, plausibility, mental state, valence, etc?

Open questions

Relating the language models' performance (BERT, XGLMxl?) with the prediction of fMRI activity

- (Pseudo)perplexity
 - Cfr. Schrimpf et al. 2021
- Semantic similarity
 - Existing similarity datasets (e.g., MultiSimLex) have limited coverage
 - We could use a multilingual lexicon (e.g., word2word), and see how a model can retrieve translations from and to each language
- XTREME
- Unsupervised evaluation
 - Closeness of embeddings in language / to the global centroid of the multilingual space